PRACTICAL: 01

Set up Python environment for AI and implement a simple reflex or goalbased agent in a toy environment.

Aim: To set up a Python environment for Artificial Intelligence and implement a simple Reflex Agent in a toy environment that senses the condition of rooms and cleans them when they are dirty.

Theory: An Artificial Intelligence (AI) agent is a system that perceives its environment and takes actions to achieve a specific objective. An agent receives information about the environment through sensors and performs actions using actuators.

Program / Code:

```
print("Welcome to my first AI Robot")
# D = Dirty # C = Clean room = ['D', 'C', 'D', 'C']
print("Room before cleaning:", room) def
clean_room(room):     for i in range(len(room)):
if room[i] == 'D':            print(f"Cleaning room at
position {i + 1}")            room[i] = 'C'         else:
          print(f"Room at position {i + 1} is already
clean")     return room room = clean_room(room)
print("Room safter cleaning:", room) print("All rooms
are clean!")
```

Output:

```
Welcome to my first AI Robot
Room before cleaning: ['D', 'C', 'D', 'C']
Cleaning room at position 1
Room at position 2 is already clean
Cleaning room at position 3
Room at position 4 is already clean
Room safter cleaning: ['C', 'C', 'C', 'C']
All rooms are clean!

=== Code Execution Successful ===
```

Conclusion: The Simple Reflex Agent was successfully implemented in Python. The agent checks the current condition of each room and performs the appropriate action—cleaning dirty rooms while leaving clean rooms unchanged. This experiment demonstrates the basic working principle of an AI agent based on condition-action rules.